INFOSYS SPRINGBOARD
INTERNSHIP 6.0

CYBER RISK ASSESSMENT & THREAT INTELLIGENCE PLATFORM

Presented by
Noor Nabeeha
Dhanush P
Sujithraa
Akshay
Eswar

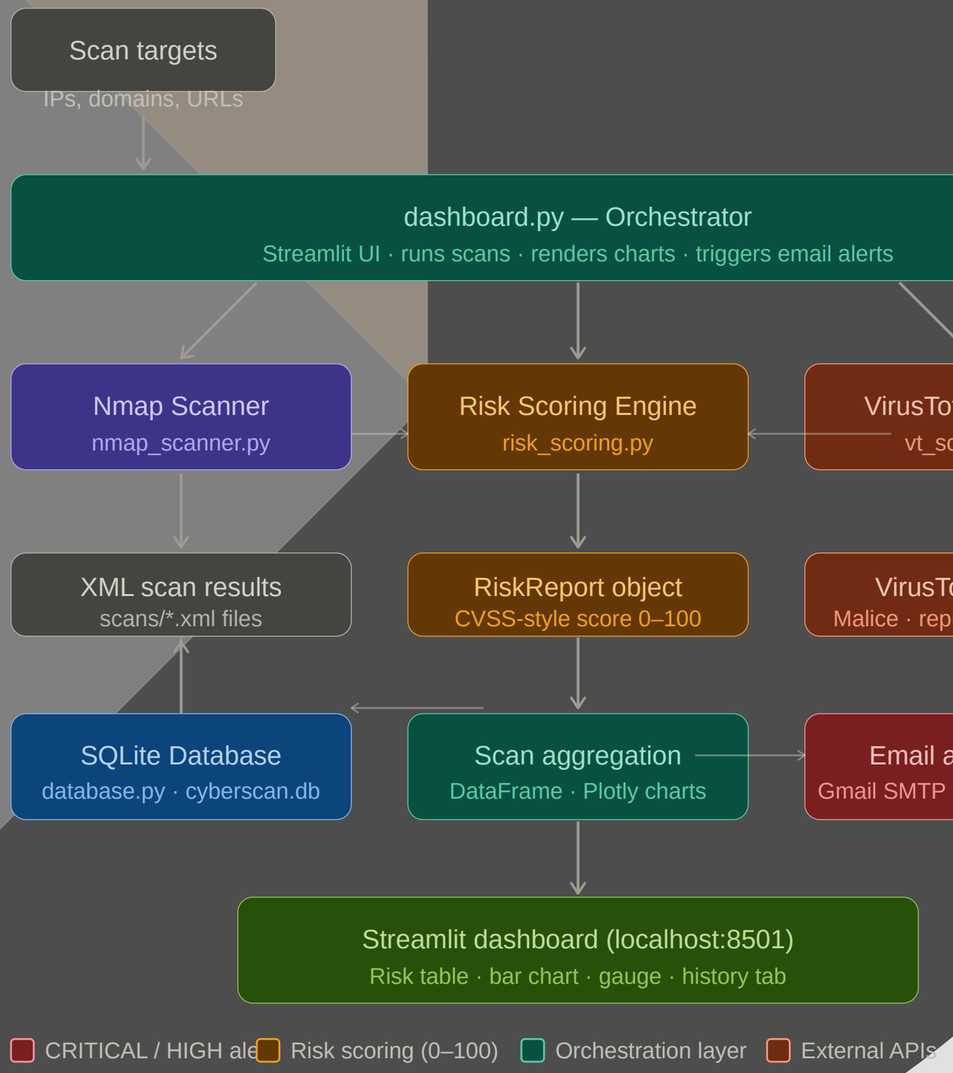
PROJECT STATEMENT:

Project Statement:

This project aims to build a Python-based platform for automating cyber risk assessments and integrating threat intelligence. The system will conduct vulnerability scans, assign risk scores, and generate security reports for enterprises. By integrating with security APIs (e.g., Shodan, VirusTotal, Nessus), the platform will provide actionable intelligence and alerts. Dashboards will visualize vulnerabilities, threat trends, and compliance status. This solution helps organizations strengthen cybersecurity posture, reduce risk exposure, and comply with security standards.

- Python-based platform for automated cyber risk assessment
- Integrates threat intelligence (Shodan, VirusTotal, Nessus)
- Performs vulnerability scanning and risk scoring
- Generates security reports and real-time alerts
- Dashboard for vulnerabilities, trends, and compliance
- Helps reduce risk and improve cybersecurity posture

PROJECT ARCHITECTURE



MODULES → SOLUTION

01

Vulnerability Scanning Engine

- Wraps Nmap as a Python subprocess (-sT -sV -O --script banner -Pn)
- Parses XML output to extract open ports, services, OS, SSL tunneling, banners
- Tags each port as HIGH / MEDIUM / LOW risk using lookup tables
- SSL on a high-risk port auto-downgrades severity (HIGH → MEDIUM)

MODULES → SOLUTION

02

Threat Intelligence Integration

- Queries VirusTotal API v3 with Base64-encoded target URLs
- Retrieves: malicious engine count, suspicious count, reputation score, last scan date
- Recursively checks up to 10 outgoing links for malicious flags
- API key loaded from .env — never hardcoded

RISK SCORING & ANALYTICS CORE

NMAP SCAN

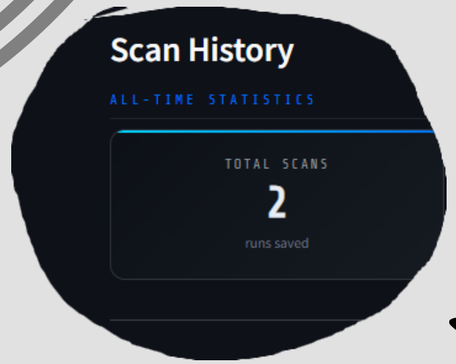
- Nmap score: +40 HIGH port, +20 MEDIUM, +5 open port, -10 SSL, +15 unpatched signal, +20 EOL OS

VT SCAN

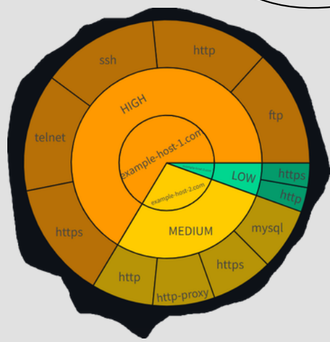
- VT score: weighted penalties for malicious hits, suspicious hits, bad reputation, malicious outlinks, stale data

- **Composite** = 55% Nmap + 45% VT, clamped 0–100
- **Severity labels:** CRITICAL (≥80) · HIGH (≥60) · MEDIUM (≥40) · LOW (≥20) · INFO

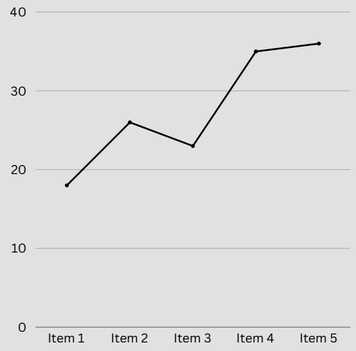
DASHBOARD

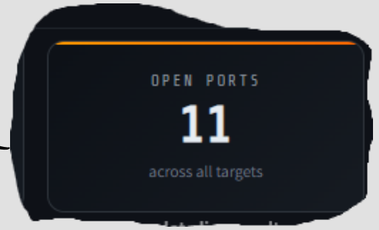


- **SQLite persistence via database.py**

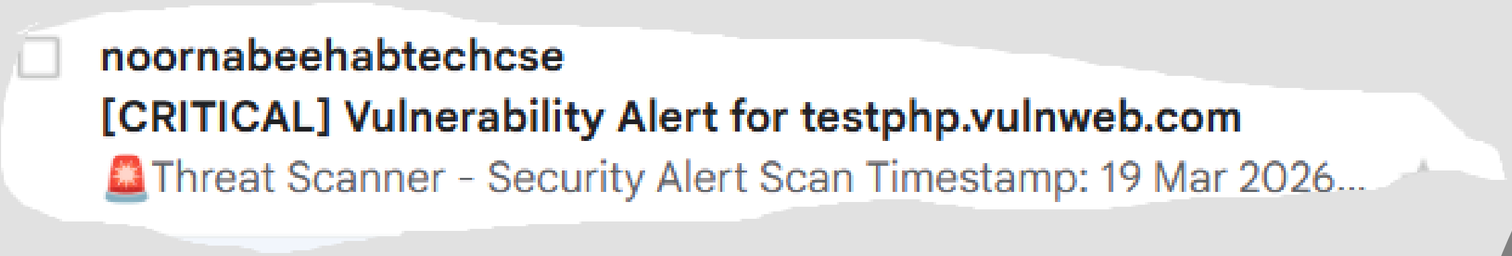


- **metric cards,**
- **Plotly charts,**
- **gauge meters per target**





- **Auto email alert (Gmail SMTP) triggered for any HIGH or CRITICAL result**



KEY BENEFITS

- Fully automated, on-demand scanning – no manual steps
- Scores are deterministic and auditable (documented weights)
- **Dual-source scoring:** network-level (Nmap) + reputation-level (VirusTotal)
- Persistent **scan history enables trend tracking** over time
- Instant **email alerts** for high-severity findings

LEARNINGS

- Python stdlib alone (subprocess, sqlite3, smtplib, xml) is enough for production-grade security tooling
- Dataclasses (PortRisk, RiskReport) make scoring logic clean and testable
- Clean module separation: scan → score → persist → present
- .env + .gitignore pattern enforces real credential hygiene

Project Structure

```
Cyber-Risk-and-Threat-Intelligence-platform/
│
├── dashboard.py                  # Main Streamlit app — entry point
├── database.py                   # SQLite persistence layer
├── .env                          # API keys & credentials (not committed)
├── requirements.txt              # Python dependencies
├── sample_test_targets.txt       # Sample scan targets
├── cyberscan_logo-removebg-preview.png  # App logo
│
├── scanners/
│   ├── __init__.py
│   ├── risk_scoring.py           # Composite risk scoring engine
│   ├── nmap_scanner/
│   │   ├── __init__.py
│   │   └── nmap_scanner.py       # Nmap scan runner & XML parser
│   └── vt_scanner/
│       ├── __init__.py
│       └── vt_scanner.py         # VirusTotal API client
│
└── scans/                        # Auto-created; stores Nmap XML output files
```

CONCLUSION

- Delivers on all 4 objectives: scan · score · visualize · alert
- Composite scoring gives one actionable number per target
- Clean architecture makes it maintainable and extensible
- With parallel scanning → ready for enterprise use